Chapter 11, Problem 13P

Problem

The Thevenin impedance of a source is **Z**Th = 120 + *j*60 Ω, while the peak Thevenin voltage is **V**Th = 165 + *j*0 V. Determine the maximum available average power from the source.

Step-by-step solution

Step 1 of 3

Consider the following data:

Thevenin's equivalent impedance, $\mathbf{Z}_{\text{Th}} = 120 + j60\ \Omega$

Thevenin's equivalent voltage, $\mathbf{V}_{\text{Th}} = 165 + j0\ \text{V}$

Step 2 of 3

The following is the load impedance for maximum power transfer:

$$\begin{aligned}\mathbf{Z}_L &= \mathbf{Z}^*_{\text{Th}} \\ &= 120 - j60\ \Omega\end{aligned}$$

Calculate the maximum rms current $|\mathbf{I}_{\text{L}}|_{\text{rms}}$ delivered to load:

$$\begin{aligned}|\mathbf{I}_{\text{L}}|_{\text{rms}} &= \left(\frac{\mathbf{V}_{\text{Th}}}{2\mathbf{Z}_{\text{Th}}}\right)\left(\frac{1}{\sqrt{2}}\right) \\ &= \left(\frac{165}{2(120)}\right)\left(\frac{1}{\sqrt{2}}\right) \\ &= 0.486\ \text{A}\end{aligned}$$

Step 3 of 3

Calculate the maximum available average power P_{avg} from the source:

$$\begin{aligned}\text{P}_{\text{avg}} &= \left(|\mathbf{I}_{\text{L}}|_{\text{rms}}\right)^2 \mathbf{Z}_{\text{Th}} \\ &= (0.486\ \text{A})^2 (120\ \Omega) \\ &= (0.486)^2 (120) \\ &= 28.359\ \text{W}\end{aligned}$$

Therefore, the maximum power P_{avg} available from the load is $\boxed{28.359\ \text{W}}$.

Comments (1)

Anonymous

faster way is just taking Pavg = Vth^2/(8*Rth)